Certified surface-saturation times for constant-flux sphere diffusion with Chen2020 parameters

(a model computation; the series is classical)

Xavier Callens (SocrateAI Lab)

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AI-assistance disclosure and audit status. This preprint, its code, its claim ledger and the audit were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **The computation was independently audited by a different AI agent instance (`AUDIT.md`, `ACCEPT_WITH_CORRECTIONS`), not by a human.** The audit's corrections, including a rounding fix to the seconds endpoints, are applied (Section 6). No human audit exists. There is no Lean in this paper.

Abstract

For diffusion in a sphere with uniform start x_0 and constant surface flux, the surface stoichiometry is $x_s = x_0 + \beta F(\tau)$ with the classical series $F(\tau) = 3\tau + \frac{1}{5} - 2\sum_n e^{-\alpha_n^2 \tau}/\alpha_n^2$, $\tan\alpha_n = \alpha_n$, $\tau = Dt/R^2$. With the negative-particle parameters printed from the Chen2020 set shipped in PyBaMM 26.10.1.1 we compute rigorous enclosures (python-flint/Arb, 300 bits) of the first time t^* at which x_s reaches x^*, for $C \in \{1, 2, 3, 4\}$, $x_0 \in \{0.05, 0.25, 0.5\}$ and $x^* \in \{1,\ 29866/33133 \approx 0.9014\}$ (24 cases; relative width $\leq 9.8 \times 10^{-14}$). The *rigorous* enclosures are exact rational bounds on τ; the seconds bounds are rounded outward after an audit found that round-to-nearest conversion had broken containment. **The certificate adds little beyond the closed-form long-time formula:** $t^* = (R^2/3D)\big[(x^* - x_0)/(C\beta) - \frac{1}{5}\big]$ up to a relative correction between 8.5×10^{-36} and 8.9×10^{-5} (below 10^{-12} in 11 of 24 cases). The value is the rigorous enclosure, an independent auditor's solver (mpmath interval arithmetic, no shared code) that reproduces it, and a model-versus-model observation (Tier [X]): PyBaMM's SPM and SPMe agree with the enclosure, but in PyBaMM's DFN the *maximum over the electrode thickness* of the surface concentration crosses 0.9014 up to 37% earlier than the certified SPM time at 1C, with a mesh dependence of about 1.5–6%, and the DFN fails to integrate at $\geq$ 3C because its state leaves its valid range. **No claim is made about faster charging, safety, lithium plating or real cells.**

What this is NOT. Not a new formula (the series is classical), not a statement about any cell, not a charging protocol, not formalised (Tier A: none). $x^* = 0.9014$ is the set's 100% state-of-charge balancing limit, *not* a saturation; only $x^* = 1$ is saturation. The start states x_0 are hypothetical. One parameter set; Chen2020 and Crank were read by neither the producer nor the auditor.

Markers. [L]: a closed form or parameter values taken as stated. [C]: rigorous or exact computation (Arb/mpmath enclosures) with the argument written down, not machine-checked by a proof assistant. [X]: float checks (finite volume, PyBaMM SPM/SPMe/DFN) and recorded absences. There is no Tier A row. Identifiers `E2S-·-nnnn` refer to `ledger/ledger.json`.

1 Problem and closed form

Model. Single-particle-model (SPM) negative particle: $c_t = Dr^{-2}(r^2 c_r)_r$ on $0 < r < R$, $c_r(0) = 0$, uniform $c(r, 0) = x_0 c_{\max}$. During charge a molar flux $j > 0$ enters the particle, so $D\, c_r(R) = +j$, with $j = I/(a\, L\, A_{\text{cell}}\, F)$, $a = 3\varepsilon_s/R$, A_{cell} = height×width and $I = C \cdot 5$ A. With $u = (c - c_0)/(\beta c_{\max})$, $\rho = r/R$, $\tau = Dt/R^2$ and $\beta = jR/(Dc_{\max})$ the problem is $u_\tau = \rho^{-2}(\rho^2 u_\rho)_\rho$ with $u_\rho(1) = 1$. The volume mean of u is 3τ; the quasi-steady

profile is $3\tau + \rho^2/2 - 3/10$; the transient is a sum of $\sin(\alpha\rho)/\rho$ with $\tan\alpha = \alpha$. Hence (derived independently by the producer and by the auditor, Crank not consulted by either)

$$x_s(\tau) = x_0 + \beta F(\tau), \qquad F(\tau) = 3\tau + \tfrac{1}{5} - 2\sum_{n\geq 1} \frac{e^{-\alpha_n^2 \tau}}{\alpha_n^2}, \qquad \alpha_n \in (n\pi,\, n\pi + \tfrac{\pi}{2}).$$

$F' = 3 + 2\sum e^{-\alpha_n^2\tau} > 0$ and $F(0) = 0$ (using $\sum \alpha_n^{-2} = 1/10$), so the first time t^* with $x_s = x^*$ is unique, $F(\tau^*) = (x^* - x_0)/\beta$. The coefficient $-2/\alpha_n^2$ at $\rho = 1$ was confirmed by the auditor by quadrature to 30 digits for $n = 1, 2, 5$. In this paper β is 0.0826678577331 per 1C (the auditor's interval evaluation gives 0.08266785773306395…); t^* depends on $(x^* - x_0)/(C\beta)$ only.

Parameters ([L]). Printed from PyBaMM 26.10.1.1 Chen2020 (`data/data/params_chen2020.json`): $D = 3.3 \times 10^{-14}$ m^2/s, $R = 5.86 \times 10^{-6}$ m, $c_{\max} = 33133$ mol/m^3, $L = 85.2 \times 10^{-6}$ m, $\varepsilon_s = 0.75$, height 0.065 m, width 1.58 m, nominal 5 A, initial concentration 29866 mol/m^3, $F = 96485.33212331001$ (scipy.constants, what `pybamm.constants` uses). The set's 100% negative stoichiometry is $29866/33133 \approx 0.9014$; secondary sources quote Chen et al. 2020 for the window 0.0279–0.9014 (the paper was not read). This is a cell-balancing limit, not a saturation. The three starts x_0 are hypothetical.

2 Certification (Tier [C])

Implementation: `code/spm_cert.py`, `code/run_cert.py` (python-flint Arb at 300 bits; parameters as exact decimals).

- **Roots.** $g(a) = \sin a - a\cos a$ has $g'(a) = a\sin a$ of constant sign on $(n\pi, n\pi + \pi/2)$; a rigorous sign change there gives exactly one root, which is the n-th positive root of $\tan a = a$ (none lies in $(n\pi - \pi/2, n\pi)$ or $(0, \pi/2)$). Widths about 10^{-42}.
- **Tail.** Since $\alpha_n \geq n\pi$, the series tail beyond N is bounded by a geometric series below 2^{-100} and added as the ball $[0, \text{tail}]$.
- **Bracketing.** τ^* is bracketed by interval bisection; a side is decided only when the F ball lies strictly on one side of the target. The result is a pair of exact rationals $[\tau_{lo}, \tau_{hi}]$.
- **Seconds.** $t_{lo} = \text{nextafter}(\text{float}(\underline{\tau_{lo} R^2/D}), -\infty)$ and $t_{hi} = \text{nextafter}(\text{float}(\overline{\tau_{hi} R^2/D}), +\infty)$ (Section 6, correction 1).

The rigorous enclosures are the exact rational τ bounds in `data/data/certified_t_star.json`; the seconds bounds are rounded outward. The maximum relative width over the 24 cases is 9.8×10^{-14} (preregistered target 10^{-9}; kill criterion (i) did not fire). Table 1 gives the mid-points.

Exact-rational verification of the seconds interval. `code/check_exact_rational.py` (log `data/logs/check_exact_rational.log.txt`) uses `fractions.Fraction`: after the fix the seconds interval contains the producer's rational τ interval times R^2/D in 24/24 cases and the auditor's independent mpmath.iv enclosure (`data/data/auditor_mpmath_iv_enclosure.json`) in 24/24. Before the fix (pre-fix output kept as `data/data/certified_t_star_prefix_roundtonearest.json`) the same check gave 2/24 and 21/24.

3 The long-time formula: what the certificate adds

For all 24 cases t^* equals the two-term formula

$$t^* = \frac{R^2}{3D}\Big[\frac{x^* - x_0}{C\beta} - \frac{1}{5}\Big]$$

up to the exponential term. The relative correction $2e^{-\alpha_1^2\tau}/(\alpha_1^2 F)$ ranges from 8.5×10^{-36} ($C = 1$, $x_0 = 0.05$, $x^* = 1$) to 8.9×10^{-5} ($C = 4$, $x_0 = 0.5$, $x^* = 0.9014$); it is below 10^{-12} in 11 of 24 cases (auditor's figures,

C	x_0	x^*	t^* (s), certified	rel. width	corr. to long-time formula
1	0.05	1	3916.716	6.5e-14	8.5e-36
1	0.05	0.9014	3502.991	7.2e-14	2.9e-32
1	0.25	1	3077.539	8.2e-14	1.3e-28
1	0.25	0.9014	2663.814	9.5e-14	4.5e-25
1	0.50	1	2028.568	6.2e-14	1.3e-19
1	0.50	0.9014	1614.844	7.8e-14	5.0e-16
2	0.05	1	1923.671	6.6e-14	1.1e-18
2	0.05	0.9014	1716.809	7.4e-14	6.6e-17
2	0.25	1	1504.083	8.4e-14	4.6e-15
2	0.25	0.9014	1297.221	9.8e-14	2.9e-13
2	0.50	1	979.598	6.5e-14	1.8e-10
2	0.50	0.9014	772.735	8.2e-14	1.3e-08
3	0.05	1	1259.323	5.1e-14	6.3e-13
3	0.05	0.9014	1121.415	5.7e-14	1.0e-11
3	0.25	1	979.598	6.5e-14	1.8e-10
3	0.25	0.9014	841.689	7.5e-14	3.0e-09
3	0.50	1	629.941	5.0e-14	2.4e-07
3	0.50	0.9014	492.035	6.4e-14	4.4e-06
4	0.05	1	927.149	6.8e-14	5.3e-10
4	0.05	0.9014	823.718	7.7e-14	4.4e-09
4	0.25	1	717.355	8.8e-14	3.9e-08
4	0.25	0.9014	613.924	5.2e-14	3.4e-07
4	0.50	1	455.117	7.0e-14	9.6e-06
4	0.50	0.9014	351.719	9.0e-14	8.9e-05

Table 1: All 24 certified cases (t^* mid-point of the outward-rounded seconds interval, 3 decimals; full rational bounds in the JSON). The last column is the relative size of the exponential correction to the long-time formula (Section 3), recomputed with mpmath (40 digits).

recomputed here: `ledger/tables_check.json`). For example $C = 1$, $x_0 = 0.05$: 3916.7158 s. So the certified series contributes mainly that correction and the rigorous enclosure; the statement "t^* depends on $(x^* - x_0)/(C\beta)$ only" follows at once (for instance 2C from $x_0 = 0.5$ equals 3C from $x_0 = 0.25$). The series itself is classical (constant surface flux in a sphere, Crank; SPM per Marquis et al. 2019); the constant 1/5 is the one of parabolic-profile SPM reductions (for example Subramanian et al. 2005, not read).

4 Controls

Positive and negative controls were run before the numbers were reported (`data/data/cert_controls.json`, `p3_checks.json`).

- **P1** short time: the relative difference of F from $2\sqrt{\tau/\pi}$ is 2.87×10^{-2} at $\tau = 10^{-3}$ and 2.81×10^{-3} at 10^{-5} (ratio 0.098, expected ≈ 0.1).
- **P2** $F(10^{-5}) = 0.003578$; the sum rule $\sum \alpha_n^{-2} = 1/10$ holds with deficit 2.527×10^{-4} at $N = 400$, inside the auditor's *two-sided* bracket $[1/(\pi^2(N + 3/2)), 1/(\pi^2 N)] = [2.5236, 2.5330] \times 10^{-4}$ (the preregistered 10^{-12} was not reached: the series needs $\sim 1/\sqrt{\tau}$ roots); $F(2) - 6 - \frac{1}{5}$ equals the first exponential term to 5×10^{-54}.
- **P3** mpmath (50 digits, independent root finding) lies inside the Arb balls at all 20 points ($\tau = 0.05$ to 3.4); a finite-volume method of lines (400 cells, scipy BDF, rtol 10^{-10}) agrees with the series to a maximum relative error 7.7×10^{-6} (Tier [X]).
- **P4** ([X]) PyBaMM SPM (initial negative concentration $x_0 c_{\max}$, current $-C \cdot 5$ A, wide voltage cut-offs): the first crossing agrees with the certified interval, maximum relative difference 4.2×10^{-4} with 20 radial points and 1.7×10^{-5} with 100, all 24 cases; the auditor reran with 200 points and finds 3.7×10^{-7} to 4.3×10^{-6} on four cases (error shrinks with the mesh). Kill criterion (ii) did not fire.

- **N1** $D \times 1.1$ gives intervals disjoint from the originals in all 24 cases. **N2** a root bracket shifted by 0.005 and a wrong interval $[4.40, 4.45]$ are not certified; a series with $\alpha_1 + 0.01$ shifts $F(10^{-5})$ by 4.4×10^{-4} (weak but discriminating, per the auditor). **N3** a reversed flux sign returns "unreachable"; the factor 2τ in place of 3τ gives $t^* = 4616$ s against 3078 s (disjoint). The preregistered second half of N3 (the P2 long-time check with 2τ must fail) was *not* run (Section 10); the 2τ versus 3τ discrimination rests on the disjoint t^* and on P4. **N4** PyBaMM discharge from $x_0 = 0.5$ for 1500 s crosses neither level.

5 SPM, SPMe and DFN in PyBaMM (Tier [X], exploratory)

SPMe. PyBaMM's SPMe equals the SPM on the negative-particle surface concentration (crossing times agree to $\sim 10^{-7}$ relative at equal mesh): the particle sees the same x-uniform current, so "t^* survives SPM to SPMe" holds by construction and is not a finding.

DFN, $x^* = 0.9014$ only (positive electrode started at 0.99 so it is not depleted; the positive start changes only the voltage). The x-averaged surface concentration of the negative electrode crosses at the certified time to 10^{-4} at 1C (three start states, plus the 100-point mesh). The *maximum over x* (the node next to the separator, 83.07 of 85.2 μm) crosses earlier: at 1C by 24% ($x_0 = 0.05$: 2667 s against 3503 s), 30% ($x_0 = 0.25$) and 37% ($x_0 = 0.5$); at 2C at 891.6 s against 1716.8 s ($x_0 = 0.05$), 574.9 against 1297.2 ($x_0 = 0.25$), 301.7 against 772.7 ($x_0 = 0.5$). The auditor reproduced the 1C, $x_0 = 0.25$ numbers bit for bit (1870.76 s for the maximum, 2663.96 s for the average); at 10 s the interfacial current near the separator is 1.39 times that near the collector. This is the classical non-uniform reaction distribution of porous-electrode theory, a real effect of the model, not a mis-definition.

Mesh dependence. The percentages are at $x_n = 20$ nodes through the electrode (the producer refined only r_n). With $x_n = 40$ the maximum crossing at 1C, $x_0 = 0.25$ moves from 1870.8 s to 1843.5 s (1.5%) and at 3C, $x_0 = 0.25$ from 206.0 s to 194.2 s (6%): the 24/30/37% figures carry roughly 1.5–6% mesh uncertainty and are *not* mesh-converged in x.

DFN failures for $C \geq 3$, and 2C beyond $1.05t^*$. The producer did not diagnose them. The auditor's diagnosis at 3C, $x_0 = 0.25$ ($x_n = 20$ and 40): the electrolyte concentration in the negative electrode falls below 1 mol/m^3 at 368 s; the local surface concentration reaches $0.9999 c_{\max}$ at about 400 s; IDA fails at 400.5–401.2 s with a negative electrolyte concentration and a surface concentration above $c_{\max}$, which makes the Chen2020 exchange current ($\propto c_e^{1/2} c_s^{1/2} (c_{\max} - c_s)^{1/2}$) NaN. The positive surface stoichiometry is 0.589, so the positive electrode is not the cause. 4C, $x_0 = 0.05$ fails the same way at 193.6 s ($c_e = -6.6 \times 10^{-4}$ mol/m^3 at 70 μm). The model state leaves its valid range; it is not a solver-option problem. The 2C failures were not diagnosed. The first failed configurations are kept in `data/logs/*first_attempt*.log.txt`. This is a model-versus-model observation under one parameter set; it says nothing about cells, charging or safety.

6 Audit and corrections applied

What the auditor did (separate AI agent instance, Claude Opus 5.5, 2026-10-31; `AUDIT.md`). Derived F, the flux sign and the factor independently; evaluated β with mpmath.iv from the decimal strings (0.0826678577330640 per 1C, equal to the producer's); recomputed all 24 enclosures with a solver sharing no code with `spm_cert.py` (interval Newton for the roots, an integral tail bound, interval Newton for τ; widths $\sim 10^{-56}$ s): every auditor τ enclosure lies inside the producer's rational interval (24/24); reran PyBaMM with its own script; read `spm_cert.py` line by line. Verdict ACCEPT_WITH_CORRECTIONS. Not verified by the auditor: Crank and Chen et al. 2020 not read, the finite-volume P3 check, the SPMe identity and N4 not rerun, the 2C DFN failure not diagnosed. No human audit.

1. **Rounding error (genuine bookkeeping error).** Seconds endpoints had been converted with round-to-nearest `float()`. In exact arithmetic the true t^* then lay below the published t_{lo} in 4 of 24 cases (by up to 1.5×10^{-13} s) and the seconds interval failed to contain the τ interval times R^2/D in nearly all cases (we measure 22 of 24 failures; the auditor reports 23). Fixed with `math.nextafter`; JSON

regenerated; pre-fix JSON, log and controls kept; all containment checks above now pass (24/24). The physical size was negligible; the word "certified" required the fix.

2. **Long-time formula** added (Section 3).
3. $x^* = 0.9014$ reworded: a balancing limit, not a saturation.
4. **DFN failure diagnosis** replaces "I did not diagnose it".
5. **Mesh dependence in** x_n of the DFN percentages added.
6. **Deviations added:** N3's second half not run; preregistration sign slip (Section 10).
7. **Novelty statement** (Section 7).
8. **P2 sum rule** made two-sided (done, `sumrule_two_sided_ok`=true).
9. **Hashes** in `result.json` regenerated (36 files).

7 Priority: this is not new

The series is classical; no new formula is claimed. The only contribution is the certified evaluation at stated parameters, the audit and the PyBaMM comparison. A quick search by the auditor (two queries, not exhaustive) found no certified-interval treatment of this saturation time; related analytic onset-time work exists (Li et al., ACS Energy Lett. 2025, an analytic lithium-plating onset model). The producer made no search. This is an absence claim recorded at Tier [X], not a priority claim.

8 The chain, with tiers

- [L]: closed form F; Chen2020 parameters as printed by PyBaMM (taken as stated); PyBaMM BSD-3-Clause is the only third-party software used (licence read in the wheel; nothing of it is vendored; the repository holds only nine printed parameter values and producer-generated outputs).
- [C]: root brackets and tail bound (Arb); τ^* enclosures and rounded seconds; the auditor's independent mpmath.iv enclosures; exact-rational containment checks; the long-time correction table.
- [X]: finite-volume P3, PyBaMM SPM/SPMe/DFN, the DFN diagnosis, the absence-of-prior-work statement, what was not run.
- Tier A: none (declared in the preregistration; the proposal's Lean lemmas and 20 core-hours of Lean were not attempted).

9 Claim ledger

`ledger/build_ledger.py` files 9 claims: 2 at Tier [L] (`E2S-L-0001..0002`), 4 at [C] (`E2S-C-0001..0004`, the headline being `-0004`) and 3 at [X] (`E2S-X-0001..0003`). There is no Tier A row. Each carries a hashed evidence blob in `ledger/evidence/`. The Elenchus gate (`tools/ledger.py` of the SocrateAI Elenchus checkout) was run with `ledger/run_gate.py`; the verbatim output is in `ledger/gate_output.txt`. The gate checks bookkeeping (tier ceilings, dependency closure) and licenses nothing about the mathematics.

10 Limitations and deviations

- **AI audit only; no human audit.** No person has read the code or the argument.
- **Not verified by the auditor:** Crank and Chen et al. 2020 not read by anyone involved (the formula was derived independently and its coefficient checked numerically; 0.9014 checked only against secondary sources); P3 finite-volume check, SPMe identity and N4 not rerun; 2C DFN failure undiagnosed.
- **Scope.** One parameter set; hypothetical x_0; SPM is known to be inaccurate at high C-rate (Marquis et al.); Butler–Volmer kinetics and OCV do not enter the particle surface concentration at imposed current. The DFN maximum-over-x result is an observation of one model at one mesh.

- **Tier [C] is the ceiling.** No Lean, no kernel check, no statement lock.
- **Deviations from the preregistration** (`prereg_E2.json`, sha256 `514d7fdd...d9269`): (1) no Lean; (2) P1/P2 evaluated at $\tau = 10^{-3}, 10^{-5}$, P2's sum rule to 2.5×10^{-4} instead of 10^{-12}; (3) N2 used a shifted bracket and an $\alpha_1 + 0.01$ series, the "$\tan a = 2a$" variant replaced by a wrong-interval probe; (4) DFN rerun with positive start 0.99 and restricted to $x^* = 0.9014$ after the first setup failed (kept in logs); DFN for $C \geq 3$ not achieved; (5) electrode height 0.065 m in this PyBaMM version ($A_{\text{cell}} = 0.1027$ m^2); positive start changed in PyBaMM runs (voltage only); (6) F from scipy.constants; (7) **N3's second part** (P2 long-time check with 2τ must fail) was not executed; (8) **preregistration sign slip**: the text "$-D\, c_r(R) = j$" has the wrong sign for $j > 0$ inward, the code uses $D\, c_r(R) = j$ (x_s increases; N3 checks this); (9) post-audit: outward rounding and two-sided sum rule.
- **Repository guards.** `pytest tests/` and the repository guards were not run for this paper.
- **Not claimed.** No battery charges faster or is safe; no lithium-plating prediction (x^* is a model proxy); models are not devices.
- Compute: cores 2 and 3 only.

11 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; the lane records (`results/proposal4/2026-10-27/E2`) were committed by the producer in commit `cf7610a`; the audit, the rounding fix, this paper and the auditor's scratch files were uncommitted working-tree files at the time of writing; nothing was pushed or uploaded. Copies of all inputs are in `data/`: `RESULT.md`, `result.json` (sha256 of every file), `prereg_E2.json`, `AUDIT.md`, `code/`, `data/` (certified JSON, pre-fix JSON, controls, PyBaMM outputs), `logs/` (the `.log` files are stored as `.log.txt`) and `audit/` (the auditor's scratch scripts and outputs; `my_cert.json` there is the auditor's run against the pre-fix floats). `data/RESULT.md` is the post-audit text. PyBaMM itself is not shipped.

```
pip install pybamm==26.10.1.1 python-flint mpmath scipy        # PyBaMM is BSD-3-Clause; not vendored here
taskset -c 2,3 python3 code/run_cert.py          # from the repo root; writes data/certified_t_star.json, cert_controls.json
python3 code/check_exact_rational.py data/certified_t_star.json data/auditor_mpmath_iv_enclosure.json
python3 code/auditor_mpmath_iv.py data/certified_t_star.json   # the auditor's independent solver (edit its output path)
python3 code/run_pybamm.py ; python3 code/run_dfn.py ; python3 code/run_dfn_early.py ; python3 code/run_p3.py
python3 -I ledger/gen_tables.py ; python3 -I ledger/build_ledger.py ; python3 -I ledger/run_gate.py
```